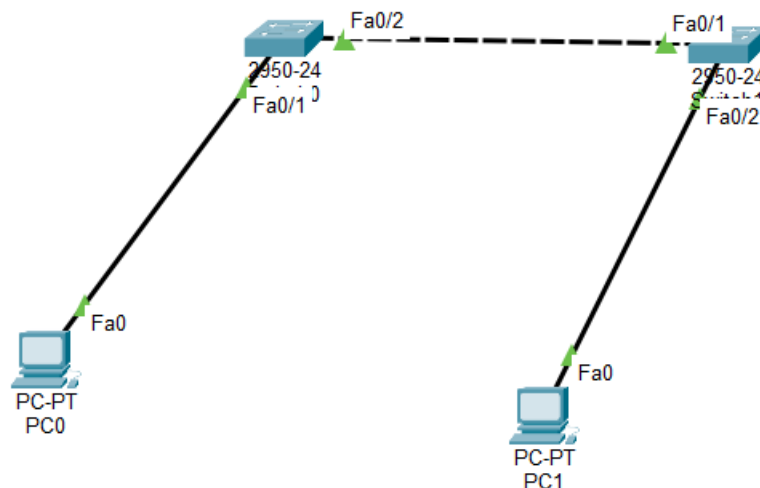


VLAN Trunking and Access Ports Theory : Trunking protocols (802.1Q, ISL).

Practical: Configure trunk ports on a switch. Task: Simulate communication between VLANs using trunk ports.

Task 1.

In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.



Switch 1 Trunk

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20

Switch#
```

Switch 2 Trunk

```
Switch#show interfaces trunk
Port      Mode           Encapsulation  Status        Native vlan
Fa0/1     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1
```

Task 2.

Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

```
Switch(config)#
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name vlan10
Switch(config-vlan)#ex
Switch(config)#vlan 20
Switch(config-vlan)#name vlan20
Switch(config-vlan)#ex
Switch(config)#vlan 30
Switch(config-vlan)#name vlan30
Switch(config-vlan)#ex
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#int Fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)%%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.

switchport
% Incomplete command.
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config)#int Fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch
FastEthernet0/1 (1).

Switch(config-if)#switchport access vlan 20
Switch(config-if)#ex
Switch(config)#int Fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
10	vlan10	active	Fa0/2
20	vlan20	active	Fa0/3
30	vlan30	active	Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

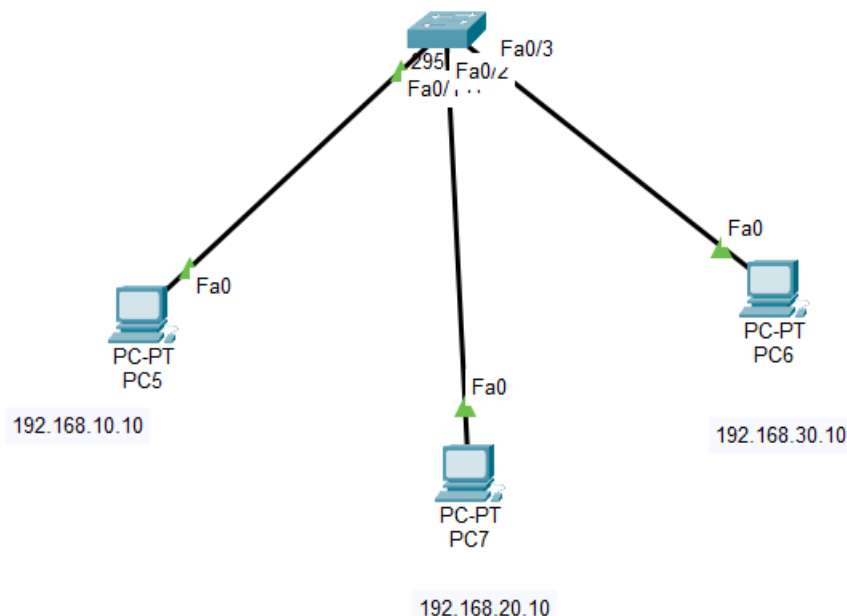
```
Switch#  
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch  
FastEthernet0/1 (1).
```

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch  
FastEthernet0/1 (1).
```

Task 3.

Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

Different network



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

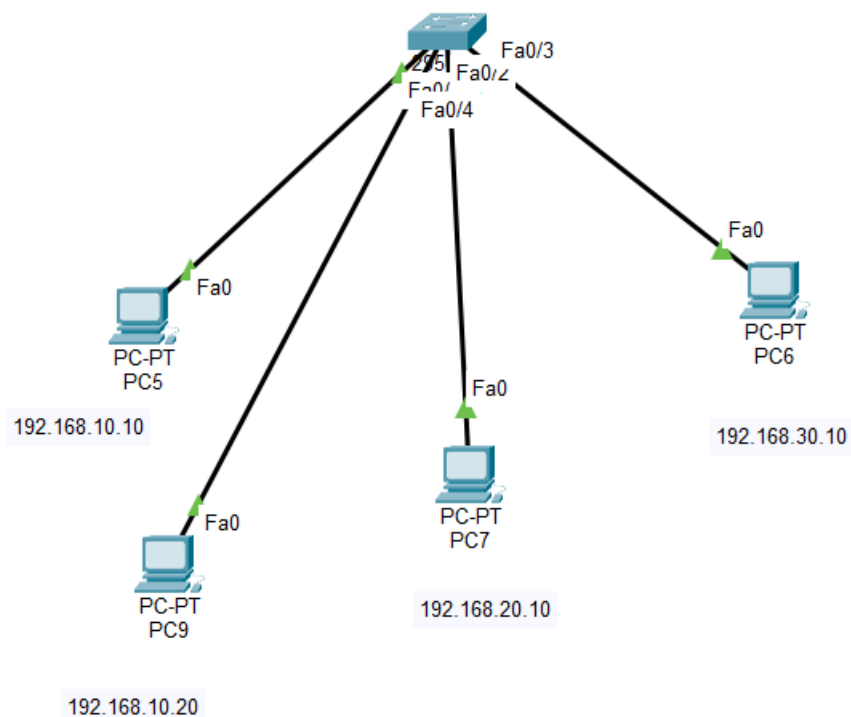
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

Same networkk



```

C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

PC	VLAN	IP Address	Mask
PC1	10	192.168.10.10	255.255.255.0
PC2	10	192.168.10.20	255.255.255.0
PC3	20	192.168.20.10	255.255.255.0
PC4	30	192.168.30.10	255.255.255.0

Task 4.

**Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

Hint: Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.**

```
Switch#show interfaces Fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: dynamic desirable
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Appliance trust: none
```

Task 5.

Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

```
Switch#show interfaces fa0/5 status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/5     Fa0/5     connected   1         a-full  a-100 10/100BaseTX
Switch#
```

```
Switch#show interfaces fa0/5 switchport
Name: Fa0/5
Switchport: Enabled
Administrative Mode: dynamic desirable
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Appliance trust: none
```

```
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Before fixing

Fa0/5:

switchport mode trunk

Result:

- Fa0/5 operates as a trunk.
- The PC is not being treated as a normal VLAN 20 access device.
- VLAN 20 connectivity may fail because the PC sends ordinary untagged frames.
- A trunk port is normally intended for switch-to-switch or other VLAN-aware device connections.

After fixing

Fa0/5:

switchport mode access

switchport access vlan 20

Result:

- Fa0/5 becomes an access port.
- The PC is assigned to VLAN 20.
- PC4 can communicate with other devices in VLAN 20.
- The VLAN 20 traffic can cross the Switch1–Switch2 trunk.

Observation: Fa0/5 was accidentally configured as a trunk instead of an access port. Since the connected PC sends normal untagged Ethernet frames, the port did not behave as a normal VLAN 20 PC connection. After changing Fa0/5 to access mode and assigning it to VLAN 20, the PC was correctly placed in VLAN 20 and communication with another VLAN 20 PC was restored.

